

I went through the **Data Consumption Requirements** section of the RFP (Pages **83–89**) and below is a complete explanation in simple words.

What is the Consumption Layer?

Think of the complete Data Lakehouse as a **large library**.

- **Data Ingestion Layer** = Librarian collecting books from different places.
- **Data Storage Layer (Lakehouse)** = Books stored neatly.
- **Data Governance Layer** = Rules about who can read which books.
- **Consumption Layer** = The people and applications that **use those books**.

So the **Consumption Layer** is the part where **business users, applications, dashboards, AI models, fraud detection systems, APIs and reports consume (use) the data stored inside the Lakehouse**.

OICL's Vision

OICL does **NOT** want every application talking directly to every other application.

Current way (Old)

INLIAS -----> HRMS

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----> SAP

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---> Dashboard

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---> Fraud System

This becomes messy.

Instead they want

Data Lakehouse

|

| | | | |

Dashboard AI Fraud APIs Reports

Everything goes through the **Lakehouse**.

This is called

Common Data Exchange Hub

Requirement 1

Common Data Exchange Hub

This is probably the **most important requirement** in the whole RFP.

The bidder has to make the Lakehouse become the **Enterprise Data Exchange Hub**.

Meaning

Every system

- publishes data
- receives data
- subscribes to data

through ONE central platform.

Example

Today

Claims System

|

|

v

Fraud System

Tomorrow

Claims System

|

v

Data Lakehouse

|
v

Fraud System

Everything passes through the Lakehouse.

Why?

Because OICL has

- INLIAS
- HRMS
- SAP
- ECM
- Investment systems
- Health Claims
- Motor claims
- Dashboards
- Fraud engine
- AI Models

If each connects separately

20 systems

Need around

190 direct connections

Impossible to maintain.

Instead

20 systems

↓

Only connect once

↓

Lakehouse

Much easier.

Data Exchange Methods Required

The solution must support ALL kinds of data movement.

API

Application asks

Give Customer Details

Returns immediately.

Batch Files

Night job

Export today's claims

Upload

Process

CDC (Change Data Capture)

Only changed data moves.

Example

Customer changed phone number.

Instead of moving

Entire Customer Table

Move only

Phone Number Changed

Very fast.

Streaming

Real-time events

Example

New Claim Registered

Immediately

Fraud Engine gets notification

Outbound Data

Lakehouse should also send data

To

- Dashboards
- Reports
- ML Models
- External Systems

Everything.

Loading Mechanisms Required

The platform should accept data using

- APIs
- SFTP
- Batch Jobs

- Database Push
- Database Pull
- Portal Upload
- Other exchange mechanisms

Meaning OICL doesn't know what partners will use.

So bidder must support all.

Single Source of Truth (SSOT)

One extremely important requirement.

Lakehouse becomes

Single Source Of Truth

Meaning

If HRMS wants policy data

Don't ask INLIAS.

Ask Lakehouse.

If Dashboard wants claims

Don't ask Claims Database.

Ask Lakehouse.

If Fraud Engine wants customer history

Don't ask multiple databases.

Ask Lakehouse.

Everything comes from one place.

No Peer-to-Peer Communication

RFP clearly says

No direct communication allowed

System A

↓

System B

Instead

System A

↓

Lakehouse

↓

System B

This becomes mandatory after implementation.

APIs Required

Bidder must create

- REST APIs
- GraphQL (or equivalent)
- Event Streams

for every consuming application.

Gold Layer

Lakehouse has

Bronze

↓

Silver

↓

Gold

Gold means

Clean

Validated

Business Ready Data

Only this should be exposed.

Event Driven Architecture

The platform should support

Kafka

or similar messaging.

Meaning

Whenever

Policy Issued

Immediately

Dashboard Updated

Fraud Updated

Analytics Updated

Notification Triggered

No waiting.

Canonical Data Model

Very important.

Every system stores customer differently.

Example

INLIAS

Cust_ID

SAP

CustomerNumber

HRMS

EmployeeCustomerID

Lakehouse converts everything into one common format.

Called

Canonical Enterprise Model.

Master Data

Must centrally manage

- Customer
- Policy
- Claim
- Agent
- Product
- Branch

Only one master copy.

Governance Requirements

Every exchange must have

- ✓ Audit Logs
- ✓ RBAC
- ✓ Authentication
- ✓ Data Lineage
- ✓ Monitoring Dashboard

No data movement without tracking.

Required Integrations

Must integrate with

- TPA
- Motor DSP
- Health Claim Documents
- INLIAS
- HRMS
- Investment Systems
- ECM
- Underwriting repositories
- Claim databases

Two-way synchronization required.

Fraud Integration

Consumption layer should

Trigger

High Risk Alert

↓

Fraud Engine

↓

Real Time Fraud Score

↓

Return Score through REST API

This is specifically required.

Requirement 2

Advanced Data Analytics

This is where the real intelligence happens.

The RFP wants the platform to become an AI-driven analytics engine.

It should support

- AI
 - ML
 - Deep Learning
 - Predictive Analytics
 - Business Intelligence
 - Risk Analytics
 - Fraud Analytics
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Analytics Required

The platform must perform

Clustering

Classification

Profiling

Pattern Recognition

Anomaly Detection

Regression

Scoring

Similarity Matching

Machine Learning

Predictive Modelling

Link Prediction

Causal Modelling

Everything listed in the RFP.

Unstructured Data

Platform should analyse

- PDFs
- Word
- Excel
- JSON
- XML
- OCR scanned documents

Not only databases.

Meaning claim documents also become searchable.

AI Features

Platform should

Learn continuously

Improve models

Forecast future

Predict claims

Detect fraud

Recommend actions

Support natural language Q&A

"What are the top fraud-prone hospitals this month?"

MLOps Required

Models should

Develop

↓

Train

↓

Deploy

↓

Monitor

↓

Retrain

Automatically.

Integrated with DevSecOps.

Business Dashboards

Must provide dashboards for

- Fraud
- Risk
- Compliance
- Data Quality

plus self-service reporting.

Analytical Tools Required

The bidder must provide tools for

- Benchmarking
- Predictive Analytics
- Prescriptive Analytics
- What-if Analysis
- Social Media Analytics
- Web Analytics
- Geolocation Analytics
- Trend Analysis
- Profit Analysis

- In-memory Analytics
 - Statistical Analytics
 - Data Mining
 - Machine Learning
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Business KPIs

The solution should calculate insurance KPIs such as:

- Gross Written Premium
 - Earned Premium
 - Claim Frequency
 - Claim Severity
 - Loss Cost
 - Incurred Claims Ratio (ICR)
 - Combined Ratio
 - Persistency Ratio
 - High-Cost Claims %
 - Policy Count
 - Lives Covered
 - Sum Insured Distribution
 - Claim Turnaround Time (TAT)
 - Underwriting Leakage
 - Renewal Ratio
 - Lapse / Churn Rate
 - Portfolio Mix
 - Anti-selection Indicators
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Predictive Models

The Lakehouse should support models for:

- Frequency prediction
- Severity prediction
- Fraud probability
- Renewal probability
- Expected loss cost

using years of policy and claims data plus external datasets where applicable. It also expects techniques such as GLM, Decision Trees, Random Forests, XGBoost, Neural Networks, LSTM/GRU, Autoencoders, BERT/LLM embeddings, K-Means, Isolation Forest, Monte Carlo Simulation, and Linear Programming.

GCC (Government Community Cloud) requirements

Although the detailed hosting section appears later in the RFP, the document consistently requires deployment on a **Government Community Cloud (GCC)**. The acronym is defined as **Government Community Cloud**.

From the hosting and deployment requirements, the key expectations are:

- The deployment model is **Private GCC**.
- The Cloud Service Provider (CSP) must be **Government of India (MeitY) empanelled** for GCC services.
- The bidder must provide **documentary proof** of GCC empanelment.
- The selected CSP must appear in the official GCC empanelment list.
- Core solution components are expected to run **compulsorily on GCC**.
- The deployment must satisfy **DPDP Act** and **IRDAI** regulatory requirements.
- The bidder is responsible for managing the cloud infrastructure and services deployed on GCC.

In simple terms, OICL is saying:

"Do not build this on any ordinary public cloud. Build it on a Government-approved cloud platform that meets Indian government security, compliance, and data-residency standards."

In the next section of the RFP, I can also explain the **Visualization, Dashboarding & Reporting, Fraud Detection & Risk Scoring, AI Workbench, and Electronic Health Claim Document Storage** requirements in the same detailed, beginner-friendly manner.

Yes. If your goal is to understand **everything related to the Consumption Layer** for technical discussions, OEM selection, solution design, or tender response, these are the pages you should study.

Page No.	Section	Importance
83–85	2.3.5.1 Common Data Exchange Hub	★★★★★ Mandatory. Explains how applications consume and exchange data through the Lakehouse.
85–92	2.3.5.2 Advanced Data Analytics	★★★★★ AI/ML, predictive analytics, business analytics, statistical analysis, MLOps, etc.
92–105	2.3.5.3 Visualization, Dashboarding & Reporting	★★★★★ BI dashboards, reports, KPIs, self-service analytics.
105–123	2.3.5.4 Fraud Detection & Risk Scoring	★★★★★ Real-time fraud analytics, risk models, anomaly detection.
123–129	2.3.5.5 AI Workbench / Data Science Platform	★★★★★ Model development, training, deployment, notebooks, MLOps.
129–130	2.3.5.6 Electronic Storage & Retrieval of Health Claim Documents	★★★ Document management, OCR, indexing, search.

These six sections together form the **Data Consumption Layer**.

Pages you should also read (very important)

The Consumption Layer cannot work independently. It depends on the sections below.

Page No.	Section	Why it matters
57–69	Data Lake / Lakehouse Setup	Understands where consumed data comes from.
69–81	Data Governance & Compliance	Data quality, metadata, catalog, lineage.

Page No.	Section	Why it matters
81–83	Data Security & Access Control	Authentication, authorization, RBAC before users consume data.
131–145	General Scope (Hosting, Infrastructure, APIs, Data Platform, Data Virtualization, Performance, AI Workbench, Governance)	Defines technical requirements supporting the consumption layer.

Pages related to GCC (Government Community Cloud)

The acronym **GCC** is defined on **Pages 17–18** as **Government Community Cloud**.

However, the **actual GCC deployment requirements** are described later in the hosting and infrastructure sections.

You should read:

Page No.	Section	Importance
131–133	Hosting & Deployment Model	★★★★★ Private GCC deployment, cloud architecture.
133–135	Storage	Storage sizing and architecture on GCC.
135–138	Architecture, Data Platform, APIs, Data Virtualization	Core platform architecture.
138–145	Performance, Governance, Security, AI Workbench, Scalability, DR	GCC operational and security requirements.
328–332	Cloud and Data Security (AMC/FMS Phase)	GCC operational security after Go-Live.
439	Data Localization & Residency Undertaking	Mandatory compliance for GCC/data residency.

If you have limited time, read these pages first

For the **Consumption Layer**:

1. **83–85** → Common Data Exchange Hub
2. **85–92** → Advanced Analytics
3. **92–105** → Dashboards & Reporting
4. **105–123** → Fraud Detection
5. **123–129** → AI Workbench
6. **129–130** → Health Claim Document Repository

For **GCC**:

1. **131–145** → Complete Hosting & Infrastructure requirements
2. **328–332** → Cloud Security
3. **439** → Data Localization & Residency

These pages contain almost everything you'll need to understand the solution architecture, prepare an OEM comparison, or respond to technical evaluation questions for this RFP.